

Reward-history expression is layered: marginal and conditional behavior must be separated

Current reward potency, neural history readout, marginal behavioral expression, and current-conditioned structure are distinct claims.

	Current reward → behavior	History → VTA DA	History → behavior	R × current conditional structure
Natural 100E/20E 11 mice	PASS Current reward changes persistence 11/11; P=.000977	PASS History changes VTA DA $\Delta R^2=.0269$; P=.00670	PASS History changes persistence $\beta=.476$, P=.00566	SUPPORTED R×current wild P=.0278 local-main P=.0122; full local×current P=.290
QE / quinine 7 mice	PASS Current QE devaluation changes persistence 7/7; P=.015625	PASS Same-current 100E history changes DA AUC $\beta=114.98$; P=.00260	PROVISIONAL Same-current duration $\beta=.818$ P=.034; alpha-family $q\approx.055-.060$	PROVISIONAL R×current wild P=.0769 6 common-support mice
VTA stimulation 13 mice	PASS Acute ON changes persistence 11/13; P=.00342	PASS Past stimulation changes later DA unique $\Delta R^2=.002524$; $P=5\times 10^{-5}$	NOT ESTABLISHED Marginal duration history P=.742 $\alpha=.03-.60$ all marginal null	LOCAL-SENSITIVE Base R×ON P=.0372 local-main adjusted P=.899; deep gate P=.154

Key update: Stim13 remains strong evidence for neural reward history, but not for a stable deep behavioral reference effect. Condition-specific duration structure is detectable in some analyses and is sensitive to recent local context; it should not be collapsed into either a simple PASS or a simple absence of all behavioral structure.